

The rhoCentralFoam solver: algorithm, numerical schemes and comparison with other OpenFOAM compressible solvers

Scope. This document describes `rhoCentralFoam` in OpenFOAM v2412 (ESI). Code paths and dictionary keywords refer to `applications/solvers/compressible/rhoCentralFoam`. Items marked ^[unverified] require a check against the v2412 source before citation.

1 Governing equations

The solver integrates the compressible Navier–Stokes equations in conservative form:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{U}) = 0, \quad (1)$$

$$\partial_t (\rho \mathbf{U}) + \nabla \cdot (\rho \mathbf{U} \otimes \mathbf{U}) + \nabla p = \nabla \cdot \boldsymbol{\tau}, \quad (2)$$

$$\partial_t (\rho E) + \nabla \cdot [\mathbf{U}(\rho E + p)] = \nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{U}) - \nabla \cdot \mathbf{q}, \quad (3)$$

with $E = e + \frac{1}{2}|\mathbf{U}|^2$, $\boldsymbol{\tau} = \mu[\nabla \mathbf{U} + (\nabla \mathbf{U})^T - \frac{2}{3}(\nabla \cdot \mathbf{U})\mathbf{I}]$ and Fourier heat flux $\mathbf{q}$. The thermophysical model must be of type `psiThermo`, with $p = \rho/\psi$ and $\psi = 1/(RT)$ for a perfect gas. The energy variable must be the sensible internal energy e . The solver stops with a fatal error if the energy variable is not e .

2 Spatial discretisation

The solver uses a cell-centred, colocated finite volume method on arbitrary polyhedral cells. Integration over a cell P of volume V_P gives the semi-discrete form

$$\frac{d\mathbf{W}_P}{dt} = -\frac{1}{V_P} \sum_{f \in \partial P} \mathbf{F}_f + \text{viscous terms}, \quad \mathbf{W} = (\rho, \rho \mathbf{U}, \rho E)^T. \quad (4)$$

The inviscid face flux $\mathbf{F}_f$ comes from the central-upwind schemes of Kurganov and Tadmor [2] and Kurganov, Noelle and Petrova [3]. Greenshields et al. [1] adapted these schemes to the polyhedral OpenFOAM framework. The adaptation needs no Riemann solver and no characteristic decomposition.

2.1 Face reconstruction

Each face f has an owner cell and a neighbour cell. The solver reconstructs two face states: the owner-side state (+) and the neighbour-side state (−). It reconstructs five fields: ρ , $\rho \mathbf{U}$, $1/\psi$, e and the speed of sound $c = \sqrt{\gamma/\psi}$. The primitive face states follow from these fields:

$$\mathbf{U}_f^\pm = \frac{(\rho \mathbf{U})_f^\pm}{\rho_f^\pm}, \quad p_f^\pm = \rho_f^\pm (1/\psi)_f^\pm, \quad \phi_f^\pm = \mathbf{U}_f^\pm \cdot \mathbf{S}_f. \quad (5)$$

The reconstruction uses the OpenFOAM limited interpolation. The solver supplies an artificial flux field equal to $+1$ for the owner side and -1 for the neighbour side. This flux sets the upwind cell for each side. The limited face value is a blend of a linear and an upwind value:

$$\varphi_f = \Psi(r) \varphi_f^{\text{lin}} + [1 - \Psi(r)] \varphi_f^{\text{up}}. \quad (6)$$

Ψ is the flux limiter. On an unstructured mesh, OpenFOAM computes the gradient ratio r from the upwind-cell gradient:

$$r = 2 \frac{\mathbf{d} \cdot (\nabla \varphi)_C}{\varphi_D - \varphi_C} - 1, \quad (7)$$

with C the upwind cell, D the downwind cell and $\mathbf{d} = \mathbf{x}_D - \mathbf{x}_C$.

The scheme for each field is set in `interpolationSchemes`. The entry names are `reconstruct(rho)`, `reconstruct(U)` and `reconstruct(T)`. The $\rho\mathbf{U}$ reconstruction uses the `U` entry. The reconstruction of $1/\psi$, e and c uses the `T` entry.

2.2 Local wave speeds

The solver computes one-sided local propagation speeds, in volumetric flux units:

$$a_f^+ = \max(\phi_f^+ + c_f^+ |\mathbf{S}_f|, \phi_f^- + c_f^- |\mathbf{S}_f|, 0), \quad (8)$$

$$a_f^- = \min(\phi_f^+ - c_f^+ |\mathbf{S}_f|, \phi_f^- - c_f^- |\mathbf{S}_f|, 0). \quad (9)$$

2.3 Numerical flux

For a transported quantity Ψ , the face flux has the form

$$F_f[\Psi] = \alpha \phi_f^+ \Psi_f^+ + (1 - \alpha) \phi_f^- \Psi_f^- + \omega_f (\Psi_f^- - \Psi_f^+). \quad (10)$$

The two schemes differ only in the weight α and the diffusive coefficient ω_f :

$$\text{Kurganov--Noelle--Petrova (Kurganov): } \alpha = \frac{a_f^+}{a_f^+ - a_f^-}, \quad \omega_f = \alpha a_f^-, \quad (11)$$

$$\text{Kurganov--Tadmor (Tadmor): } \alpha = \frac{1}{2}, \quad \omega_f = -\frac{1}{2} \max(|a_f^+|, |a_f^-|).$$

The keyword `fluxScheme` in `fvSchemes` selects the scheme. The default is `Kurganov`.

The KNP form uses two one-sided speeds. It therefore adds less numerical diffusion than the KT form, which uses one symmetric speed. Both forms reduce to a Rusanov-type flux of first order when $\Psi(r) = 0$.

The three conservative fluxes are:

$$F_f^\rho = F_f[\rho], \quad (12)$$

$$\mathbf{F}_f^{\rho\mathbf{U}} = F_f[\rho\mathbf{U}] + [\alpha p_f^+ + (1 - \alpha) p_f^-] \mathbf{S}_f, \quad (13)$$

$$F_f^{\rho E} = F_f[\rho H] + \omega_f (p_f^+ - p_f^-), \quad (14)$$

with $H = E + p/\rho$. The last term in $F_f^{\rho E}$ changes the diffusive part from ρH to ρE . The diffusion thus acts on the conserved variable.

3 Time integration and solution sequence

The solver is transient and explicit for the inviscid terms. It uses forward Euler in time, so the scheme is first order in time. For viscous flow, an implicit correction step adds the diffusion terms. This operator split removes the viscous time-step limit. Only the convective limit remains.

One time step has the sequence that follows.

1. Reconstruct ρ , $\rho\mathbf{U}$, $1/\psi$, e and c on each face. Compute $a_f^\pm$, α , ω_f and the fluxes.
2. Compute the Courant number and adjust Δt (Section 3.1).
3. Solve the density equation explicitly:

$$\rho^{n+1} = \rho^n - \frac{\Delta t}{V_P} \sum_f F_f^\rho.$$

4. Solve the momentum equation explicitly (inviscid predictor):

$$(\rho\mathbf{U})^* = (\rho\mathbf{U})^n - \frac{\Delta t}{V_P} \sum_f \mathbf{F}_f^{\rho\mathbf{U}}, \quad \mathbf{U}^* = (\rho\mathbf{U})^* / \rho^{n+1}.$$

5. For viscous flow, solve the implicit momentum correction:

$$\rho^{n+1} \frac{\mathbf{U}^{n+1} - \mathbf{U}^*}{\Delta t} - \nabla \cdot (\mu_{\text{eff}} \nabla \mathbf{U}^{n+1}) - \nabla \cdot \boldsymbol{\tau}_{\text{MC}} = 0,$$

with $\boldsymbol{\tau}_{\text{MC}} = \mu_{\text{eff}} \text{dev}_2[(\nabla \mathbf{U})^T]$ explicit. Then set $(\rho\mathbf{U})^{n+1} = \rho^{n+1} \mathbf{U}^{n+1}$.

6. Solve the energy equation explicitly. The equation includes the viscous work $\nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{U})$ as an explicit term:

$$(\rho E)^* = (\rho E)^n - \frac{\Delta t}{V_P} \sum_f \left(F_f^{\rho E} - (\boldsymbol{\tau} \cdot \mathbf{U})_f \cdot \mathbf{S}_f \right).$$

Compute $e^* = (\rho E)^* / \rho^{n+1} - \frac{1}{2} |\mathbf{U}^{n+1}|^2$ and update the thermophysical model.

7. For viscous flow, solve the implicit heat-conduction correction:

$$\rho^{n+1} \frac{e^{n+1} - e^*}{\Delta t} - \nabla \cdot (\alpha_{\text{eff}} \nabla e^{n+1}) = 0.$$

Update the thermophysical model again and set $(\rho E)^{n+1} = \rho^{n+1} (e^{n+1} + \frac{1}{2} |\mathbf{U}^{n+1}|^2)$.

8. Update the pressure, $p = \rho/\psi$, and correct the turbulence model.

The keyword `inviscid` in `thermophysicalProperties` or the transport model removes steps 5 and 7 when $\mu = 0$. Check that α_{eff} is the diffusivity of e (κ/c_v), not of h (κ/c_p)^[unverified].

3.1 Time-step control

The solver defines the Courant number from the maximum flux coefficient on each face:

$$\text{Co} = \frac{\Delta t}{2} \max_P \left(\frac{1}{V_P} \sum_{f \in \partial P} \max(|\alpha \phi_f^+ - \omega_f|, |(1 - \alpha) \phi_f^- + \omega_f|) \right). \quad (15)$$

Appendix A.8 derives this definition. With `adjustTimeStep yes`, the solver sets Δt so that $\text{Co} \leq \text{maxCo}$. The distributed tutorials use `maxCo = 0.2`^[unverified]. The time step is global; the solver has no local time stepping^[unverified].

4 Flux limiters

Table 1 lists the TVD limiters in OpenFOAM that are applicable to the reconstruction. Sweby [4] gives the second-order TVD region: $r \leq \Psi(r) \leq \min(2r, 2)$ for $0 \leq r \leq 1$ and $1 \leq \Psi(r) \leq \min(r, 2)$ for $r \geq 1$. For vector fields, use the `V` variants (for example `vanLeerV`). These variants calculate one limiter from the direction of maximum change. This removes the dependence of the limiter on the coordinate system.

Table 1: TVD limiters available for `reconstruct(...)` entries.

Keyword	$\Psi(r), r > 0$	Properties
<code>Minmod</code>	$\min(r, 1)$	Lower bound of the TVD region. Most diffusive. Robust for start-up and strong shocks.
<code>vanLeer</code>	$\frac{r + r }{1 + r }$	Smooth and symmetric. Sharp shocks with moderate clipping of extrema. General default for shock-dominated flow.
<code>vanAlbada</code>	$\frac{r^2 + r}{r^2 + 1}$	Differentiable in r . Less abrupt switching. Better residual convergence to a steady state.
<code>MUSCL</code>	$\max[0, \min(2r, \frac{1+r}{2}, 2)]$	Monotonised central. Less diffusive than <code>vanLeer</code> . Can oscillate on coarse meshes.
<code>limitedLinear k</code>	$\max[0, \min(2r/k, 1)]$	$k \in (0, 1]$; $k = 1$ is the most limiting setting. Linear at $\Psi = 1$.
<code>SuperBee</code>	$\max[0, \min(2r, 1), \min(r, 2)]$	Upper bound of the TVD region. Compressive. Steepens smooth waves into steps.

4.1 Recommended combinations

Unsteady, shock-dominated flow (shock tube, forward step, blunt bodies):

```
fluxScheme Kurganov;
interpolationSchemes
{
    default linear;
    reconstruct(rho) vanLeer;
    reconstruct(U) vanLeerV;
    reconstruct(T) vanLeer;
}
```

Steady transonic aerofoil (convergence on C_D): use `vanAlbada` for `rho` and `T`, and `vanAlbadaV` for `U`. The differentiable limiter reduces the limit cycles of the residual near the shock. Keep `fluxScheme Kurganov`.

Start-up from a uniform field: use `Minmod` for the first transient, then change to the production limiter.

Schemes to avoid. Do not use `linear` reconstruction near shocks: it is not TVD and produces oscillations. Do not use `upwind`: it reduces the scheme to first order in space. Do not use `SuperBee` for viscous, turbulent or smooth flow: it distorts smooth profiles. Do not use `QUICK` or `cubic` for reconstruction: they are not TVD.

Other entries. Use `Gauss linear` for `gradSchemes`; the limiter receives this gradient through r . Use `Gauss linear corrected` for `laplacianSchemes` on orthogonal meshes. On meshes with non-orthogonality above approximately 70° , use `limited corrected 0.33`. Use `Euler` for `ddtSchemes`.

5 Comparison with other OpenFOAM compressible solvers

OpenFOAM v2412 contains one density-based solver, `rhoCentralFoam`. The other compressible solvers are pressure-based. They solve a pressure equation obtained from continuity and the

equation of state. The **transonic** option adds a convective term $\nabla \cdot (\psi \phi p)$ to that equation. This term gives the pressure equation the hyperbolic character necessary for shocks.

Table 2: Compressible solvers relevant to external aerodynamics.

Solver	Formulation	Time	Coupling	Regime and use
rhoCentralFoam	density-based	explicit, transient	none (coupled flux)	Supersonic and strongly transonic flow; sharp shocks; small Δt .
rhoPimpleFoam	pressure-based	implicit, transient	PIMPLE	All speeds with transonic ; large Courant numbers; local time stepping with localEuler .
rhoSimpleFoam	pressure-based	steady	SIM- PLE/SIM- PLEC	Steady subsonic and weakly transonic flow.
sonicFoam	pressure-based	implicit, transient	PISO	Transient transonic and supersonic flow; superseded by rhoPimpleFoam ^[unverified] .
shockFluid	density-based	explicit, transient	none	OpenFOAM Foundation v11+ module; port of rhoCentralFoam ^[unverified] .
HiSA	density-based	implicit, pseudo-time	coupled	Third party. AUSM ⁺ -up and HLLC fluxes; steady and transient aerodynamics ^[unverified] .

5.1 Main differences

Shock resolution. The inviscid fluxes of ρ , $\rho \mathbf{U}$ and ρE are in conservation form. By the Lax–Wendroff theorem [6], a converged solution therefore satisfies the Rankine–Hugoniot conditions. The viscous correction steps keep this property (Appendix A.7). Pressure-based solvers with **transonic** capture shocks, but they smear them over more cells. They can also give small errors in the jump conditions for strong shocks.

Time step and cost. **rhoCentralFoam** needs Co below one, based on $|u| + c$. On boundary-layer meshes the smallest cell sets the global Δt . **rhoPimpleFoam** with **localEuler** and **rhoSimpleFoam** do not have this limit. For a steady result, they usually need less wall-clock time.

Low Mach number. The KT and KNP diffusion scales with $|u| + c$. When $M \rightarrow 0$, this diffusion becomes large relative to the convective flux. The solver has no low-Mach preconditioning. Accuracy decreases at low Mach number, and the acoustic time-step limit increases the cost. Pressure-based solvers do not have this problem.

Contact discontinuities. The central-upwind flux does not resolve the contact wave separately. Contacts and shear layers are therefore more diffuse than with an HLLC flux.

Temporal accuracy. The forward Euler step is first order in time. For unsteady results, control the temporal error with a small Co or with a time-step convergence study.

A Full development of the algorithm

This appendix derives each step of Section 3 from the governing equations. Code variable names refer to `rhoCentralFoam.C` in OpenFOAM v2412. Table 3 maps the symbols to the code variables.

A.1 Finite-volume semi-discretisation

Write the governing equations as one system:

$$\partial_t \mathbf{W} + \nabla \cdot \mathbb{F}^c(\mathbf{W}) = \nabla \cdot \mathbb{F}^v(\mathbf{W}, \nabla \mathbf{W}), \quad (16)$$

with

$$\mathbf{W} = \begin{pmatrix} \rho \\ \rho \mathbf{U} \\ \rho E \end{pmatrix}, \quad \mathbb{F}^c = \begin{pmatrix} \rho \mathbf{U} \\ \rho \mathbf{U} \otimes \mathbf{U} + p \mathbf{I} \\ \mathbf{U}(\rho E + p) \end{pmatrix}, \quad \mathbb{F}^v = \begin{pmatrix} 0 \\ \boldsymbol{\tau} \\ \boldsymbol{\tau} \cdot \mathbf{U} - \mathbf{q} \end{pmatrix}. \quad (17)$$

Integrate over a cell P and apply the divergence theorem:

$$\frac{d}{dt} \int_{V_P} \mathbf{W} dV + \oint_{\partial V_P} \mathbb{F}^c \cdot \mathbf{n} dS = \oint_{\partial V_P} \mathbb{F}^v \cdot \mathbf{n} dS. \quad (18)$$

Define the cell average $\mathbf{W}_P = V_P^{-1} \int_{V_P} \mathbf{W} dV$. Replace each face integral by the midpoint rule, which is second-order accurate:

$$V_P \frac{d\mathbf{W}_P}{dt} = - \sum_{f \in \partial P} s_{P,f} \mathbf{F}_f^c + \sum_{f \in \partial P} s_{P,f} \mathbf{F}_f^v. \quad (19)$$

The face area vector $\mathbf{S}_f$ points from the owner cell to the neighbour cell. The sign $s_{P,f}$ is +1 if P is the owner of f and -1 if P is the neighbour. Each internal face flux therefore occurs in two cells with opposite signs. Section A.7 uses this property.

A.2 Central-upwind flux in one dimension

Consider the scalar law $u_t + f(u)_x = 0$ on a uniform grid with cell width Δx . The reconstruction (Section A.4) gives two values at the interface $x_{j+1/2}$: u^L from cell j and u^R from cell $j+1$. The discontinuity at $x_{j+1/2}$ generates waves. Their speeds are bounded by

$$a^+ = \max(\lambda_{\max}(u^L), \lambda_{\max}(u^R), 0), \quad a^- = \min(\lambda_{\min}(u^L), \lambda_{\min}(u^R), 0). \quad (20)$$

For the Euler equations, the eigenvalues in the face-normal direction are $u_n - c$, u_n and $u_n + c$. The derivation of Kurganov, Noelle and Petrova [3] has four steps.

Step 1: average over the wave fan. During a time Δt , all waves stay in the interval $[x_{j+1/2} + a^- \Delta t, x_{j+1/2} + a^+ \Delta t]$. Outside this interval, the solution near the interface is u^L on the left and u^R on the right, to $O(\Delta t)$. Integrate the conservation law over the interval and over $[t^n, t^n + \Delta t]$. The average of the solution over the fan at $t^n + \Delta t$ is

$$w_{j+1/2} = \frac{a^+ u^R - a^- u^L - [f(u^R) - f(u^L)]}{a^+ - a^-} + O(\Delta t). \quad (21)$$

Step 2: evolve the smooth part. The region between two adjacent fans, $[x_{j-1/2} + a_{j-1/2}^+ \Delta t, x_{j+1/2} + a_{j+1/2}^- \Delta t]$, contains no discontinuity. Its integral changes by the exact fluxes at its two edges, $f(u_{j-1/2}^R)$ and $f(u_{j+1/2}^L)$.

Step 3: project onto the cell. Cell j contains three parts: a strip of width $a_{j-1/2}^+ \Delta t$ from the left fan, a strip of width $-a_{j+1/2}^- \Delta t$ from the right fan, and the smooth part. Add the integrals of the three parts:

$$\Delta x \bar{u}_j^{n+1} = \Delta x \bar{u}_j^n + a_{j-1/2}^+ \Delta t (w_{j-1/2} - u_{j-1/2}^R) - a_{j+1/2}^- \Delta t (w_{j+1/2} - u_{j+1/2}^L) - \Delta t [f(u_{j+1/2}^L) - f(u_{j-1/2}^R)] + O(\Delta t^2). \quad (22)$$

The terms $-a_{j-1/2}^+ \Delta t u_{j-1/2}^R$ and $+a_{j+1/2}^- \Delta t u_{j+1/2}^L$ remove the two strips from the initial integral of the smooth part.

Step 4: take the semi-discrete limit. Divide by $\Delta x \Delta t$ and let $\Delta t \rightarrow 0$:

$$\frac{d\bar{u}_j}{dt} = -\frac{H_{j+1/2} - H_{j-1/2}}{\Delta x}, \quad (23)$$

with

$$H_{j+1/2} = f(u^L) + a^-(w - u^L) = f(u^R) + a^+(w - u^R). \quad (24)$$

Substitute Eq. (21) into either expression. Both give the same flux:

$$H_{j+1/2} = \frac{a^+ f(u^L) - a^- f(u^R)}{a^+ - a^-} + \frac{a^+ a^-}{a^+ - a^-} (u^R - u^L). \quad (25)$$

The equality of the two expressions shows that the scheme is in conservation form: the flux that leaves cell j at $x_{j+1/2}$ is the flux that enters cell $j+1$.

Special cases.

- Eq. (25) is identical to the HLL flux of Harten, Lax and van Leer [5], with wave-speed estimates $a^\pm$. The HLL flux uses two waves and has no contact wave. This is the reason why the scheme diffuses contact discontinuities and shear layers.
- Set $a^+ = -a^- = a = \max(|a^+|, |a^-|)$. Eq. (25) becomes the Kurganov–Tadmor flux [2]: $H = \frac{1}{2} [f(u^L) + f(u^R)] - \frac{a}{2} (u^R - u^L)$. This is the Rusanov (local Lax–Friedrichs) flux applied to the reconstructed states.
- If $a^- = 0$ (supersonic flow from left to right), $H = f(u^L)$. The flux is fully upwind.

A.3 Extension to polyhedral faces

OpenFOAM applies Eq. (25) along the normal of each face. It uses volumetric fluxes, with units $\text{m}^3 \text{s}^{-1}$, in place of speeds. The owner side is the left state, $L \equiv +$. The neighbour side is the right state, $R \equiv -$. Multiply all speeds by $|\mathbf{S}_f|$:

$$\phi_f^\pm = \mathbf{U}_f^\pm \cdot \mathbf{S}_f, \quad (26)$$

$$a_f^+ = \max(\phi_f^+ + c_f^+ |\mathbf{S}_f|, \phi_f^- + c_f^- |\mathbf{S}_f|, 0), \quad (27)$$

$$a_f^- = \min(\phi_f^+ - c_f^+ |\mathbf{S}_f|, \phi_f^- - c_f^- |\mathbf{S}_f|, 0). \quad (28)$$

Define

$$\alpha = \frac{a_f^+}{a_f^+ - a_f^-}, \quad 1 - \alpha = \frac{-a_f^-}{a_f^+ - a_f^-}, \quad \omega_f = \frac{a_f^+ a_f^-}{a_f^+ - a_f^-} = \alpha a_f^-. \quad (29)$$

For a convective flux $\phi \Psi$, Eq. (25) becomes Eq. (10):

$$F_f[\Psi] = \alpha \phi_f^+ \Psi_f^+ + (1 - \alpha) \phi_f^- \Psi_f^- + \omega_f (\Psi_f^- - \Psi_f^+). \quad (30)$$

The code groups the terms by face state:

$$F_f[\Psi] = \underbrace{(\alpha \phi_f^+ - \omega_f)}_{\text{aphiv_pos}} \Psi_f^+ + \underbrace{((1 - \alpha) \phi_f^- + \omega_f)}_{\text{aphiv_neg}} \Psi_f^-. \quad (31)$$

The flux is computed once per face for each transported quantity. The two coefficients in Eq. (31) are the same for all quantities.

This extension is face-by-face. It does not solve a multidimensional wave problem. The jumps of the tangential velocity are diffused with the same coefficient ω_f as the other jumps.

Table 3: Symbols and the corresponding variables in `rhoCentralFoam.C`.

Symbol	Code variable	Meaning
$\phi_f^\pm$	<code>phiv_pos</code> , <code>phiv_neg</code>	Face volumetric flux, each side
$c_f^\pm \mathbf{S}_f $	<code>cSf_pos</code> , <code>cSf_neg</code>	Acoustic volumetric flux, each side
a_f^+ , a_f^-	<code>ap</code> , <code>am</code>	One-sided local speeds
α , $1 - \alpha$	<code>a_pos</code> , <code>a_neg</code>	Weights
ω_f	<code>aSf</code>	Diffusive coefficient
$\alpha \phi_f^+ - \omega_f$	<code>aphiv_pos</code>	Owner-side coefficient
$(1 - \alpha) \phi_f^- + \omega_f$	<code>aphiv_neg</code>	Neighbour-side coefficient
F_f^ρ , $\mathbf{F}_f^{\rho\mathbf{U}}$, $F_f^{\rho E}$	<code>phi</code> , <code>phiUp</code> , <code>phiEp</code>	Inviscid face fluxes
$\boldsymbol{\tau}_{\text{MC}}$	<code>tauMC</code>	Explicit part of the stress
$(\boldsymbol{\tau} \cdot \mathbf{U})_f \cdot \mathbf{S}_f$	<code>sigmaDotU</code>	Viscous work flux

A.4 Limited reconstruction

For the owner side, the upwind cell is $C = P$ (owner) and the downwind cell is $D = N$ (neighbour). For the neighbour side, the roles change: $C = N$ and $D = P$. OpenFOAM writes the limited face value as a weighted sum:

$$\varphi_f = w_f \varphi_C + (1 - w_f) \varphi_D, \quad w_f = \Psi(r) w_f^{\text{lin}} + [1 - \Psi(r)] \cdot 1, \quad (32)$$

with w_f^{lin} the linear-interpolation weight of cell C . On a uniform mesh, $w_f^{\text{lin}} = \frac{1}{2}$ and

$$\varphi_f = \varphi_C + \frac{1}{2} \Psi(r) (\varphi_D - \varphi_C). \quad (33)$$

This is the limited-slope (MUSCL) reconstruction. $\Psi = 0$ gives the first-order upwind value. $\Psi = 1$ gives the linear value.

The ratio r uses the gradient in cell C :

$$r = 2 \frac{\mathbf{d} \cdot (\nabla \varphi)_C}{\varphi_D - \varphi_C} - 1, \quad \mathbf{d} = \mathbf{x}_D - \mathbf{x}_C. \quad (34)$$

Check on a uniform 1D mesh with upstream cell U : the central gradient gives $\mathbf{d} \cdot (\nabla \varphi)_C = \frac{1}{2}(\varphi_D - \varphi_U)$, therefore

$$r = \frac{\varphi_D - \varphi_U}{\varphi_D - \varphi_C} - 1 = \frac{\varphi_C - \varphi_U}{\varphi_D - \varphi_C}. \quad (35)$$

This is the classical ratio of consecutive gradients [4]. On an unstructured mesh, the cell gradient replaces the missing upstream cell.

A.5 Assembly of the inviscid face fluxes

Mass. The physical flux is $\rho \phi$. Therefore $F_f^\rho = F_f[\rho]$.

Momentum. The physical flux is $\rho \mathbf{U} \phi + p \mathbf{S}_f$. Apply Eq. (25) to the full flux. The convective part gives $F_f[\rho \mathbf{U}]$, which contains the diffusive term $\omega_f((\rho \mathbf{U})^- - (\rho \mathbf{U})^+)$. The pressure part gives only the weighted average, because the diffusive term already acts on $\rho \mathbf{U}$:

$$\mathbf{F}_f^{\rho \mathbf{U}} = F_f[\rho \mathbf{U}] + [\alpha p_f^+ + (1 - \alpha) p_f^-] \mathbf{S}_f. \quad (36)$$

Energy. The physical flux is $\mathbf{U} \cdot \mathbf{S}_f (\rho E + p) = \phi \rho H$, with $\rho H = \rho E + p$. Eq. (25) requires the diffusive term to act on the conserved variable ρE , not on ρH . $F_f[\rho H]$ contains the term $\omega_f((\rho H)^- - (\rho H)^+)$. Use $\rho H = \rho E + p$:

$$\omega_f((\rho E)^- - (\rho E)^+) = \omega_f((\rho H)^- - (\rho H)^+) + \omega_f(p_f^+ - p_f^-). \quad (37)$$

Therefore

$$F_f^{\rho E} = F_f[\rho H] + \omega_f(p_f^+ - p_f^-), \quad (\rho H)_f^\pm = \rho_f^\pm \left(e_f^\pm + \frac{1}{2} |\mathbf{U}_f^\pm|^2 \right) + p_f^\pm. \quad (38)$$

A.6 Time integration by operator splitting

Write Eq. (19) as

$$\frac{d\mathbf{W}}{dt} = \mathcal{A}(\mathbf{W}) + \mathcal{D}(\mathbf{W}), \quad (39)$$

with $\mathcal{A}$ the inviscid operator and $\mathcal{D}$ the viscous operator. The solver uses first-order (Lie) splitting:

$$\mathbf{W}^* = \mathbf{W}^n + \Delta t \mathcal{A}(\mathbf{W}^n) \quad (\text{explicit forward Euler}), \quad (40)$$

$$\mathbf{W}^{n+1} = \mathbf{W}^* + \Delta t \mathcal{D}(\mathbf{W}^{n+1}) \quad (\text{implicit backward Euler, partly linearised}). \quad (41)$$

The local error of each step is $O(\Delta t^2)$. The global error is $O(\Delta t)$. Backward Euler is unconditionally stable for linear diffusion. The viscous stability limit $\Delta t \lesssim \rho \Delta x^2 / \mu$ therefore does not apply.

Identity for the correction steps. The code writes each correction as `fvm::ddt(rho,X) - fvc::ddt(rho,X)`. The implicit term is $(\rho^{n+1} X^{n+1} - \rho^n X^n) / \Delta t$. The explicit term uses the current fields, $(\rho^{n+1} X^* - \rho^n X^n) / \Delta t$. The difference is

$$\text{fvm::ddt}(\text{rho}, X) - \text{fvc::ddt}(\text{rho}, X) = \rho^{n+1} \frac{X^{n+1} - X^*}{\Delta t}. \quad (42)$$

The term $\rho^n X^n$ cancels. The correction step thus starts from the predictor value X^* .

Mass. The mass equation has no viscous term:

$$\rho_P^{n+1} = \rho_P^n - \frac{\Delta t}{V_P} \sum_f s_{P,f} F_f^\rho. \quad (43)$$

Momentum predictor.

$$(\rho \mathbf{U})_P^* = (\rho \mathbf{U})_P^n - \frac{\Delta t}{V_P} \sum_f s_{P,f} \mathbf{F}_f^{\rho \mathbf{U}}, \quad \mathbf{U}_P^* = (\rho \mathbf{U})_P^* / \rho_P^{n+1}. \quad (44)$$

The division uses ρ^{n+1} , because the density is already updated.

Momentum correction. The viscous stress is $\boldsymbol{\tau} = \mu[\nabla\mathbf{U} + (\nabla\mathbf{U})^T - \frac{2}{3}(\nabla \cdot \mathbf{U})\mathbf{I}]$. Define $\text{dev}_2(\mathbf{A}) = \mathbf{A} - \frac{2}{3}\text{tr}(\mathbf{A})\mathbf{I}$. Because $\text{tr}[(\nabla\mathbf{U})^T] = \nabla \cdot \mathbf{U}$,

$$\boldsymbol{\tau} = \mu \nabla\mathbf{U} + \underbrace{\mu \text{dev}_2[(\nabla\mathbf{U})^T]}_{\boldsymbol{\tau}_{\text{MC}}}. \quad (45)$$

The first term contains the second derivatives along each direction. It gives a symmetric, diagonally dominant matrix, and the solver treats it implicitly. The second term contains the cross derivatives. The solver treats it explicitly. For constant μ and $\nabla \cdot \mathbf{U} = 0$, $\nabla \cdot \boldsymbol{\tau}_{\text{MC}} = 0$. With Eq. (42), the correction is

$$\rho^{n+1} \frac{\mathbf{U}^{n+1} - \mathbf{U}^*}{\Delta t} = \nabla \cdot (\mu_{\text{eff}} \nabla \mathbf{U}^{n+1}) + \nabla \cdot \boldsymbol{\tau}_{\text{MC}}, \quad (\rho\mathbf{U})^{n+1} = \rho^{n+1} \mathbf{U}^{n+1}. \quad (46)$$

Energy predictor. The total-energy equation contains the viscous work $\nabla \cdot (\boldsymbol{\tau} \cdot \mathbf{U})$. The solver adds it explicitly to the inviscid flux:

$$(\rho E)_P^* = (\rho E)_P^n - \frac{\Delta t}{V_P} \sum_f s_{P,f} \left[F_f^{\rho E} - (\boldsymbol{\tau} \cdot \mathbf{U})_f \cdot \mathbf{S}_f \right]. \quad (47)$$

The face value of the viscous work is^[unverified]

$$(\boldsymbol{\tau} \cdot \mathbf{U})_f \cdot \mathbf{S}_f \approx \left[\mu_{\text{eff},f} |\mathbf{S}_f| (\partial_n \mathbf{U})_f + \mathbf{S}_f \cdot (\boldsymbol{\tau}_{\text{MC}})_f \right] \cdot \left[\alpha \mathbf{U}_f^+ + (1 - \alpha) \mathbf{U}_f^- \right], \quad (48)$$

with $(\partial_n \mathbf{U})_f$ the face-normal gradient (`snGrad`). The internal energy follows from the predicted total energy and the corrected velocity:

$$e^* = \frac{(\rho E)^*}{\rho^{n+1}} - \frac{1}{2} |\mathbf{U}^{n+1}|^2. \quad (49)$$

The momentum correction decreases the kinetic energy. The viscous work in $(\rho E)^*$ does not decrease the total energy. The difference goes into e^* . This is the discrete form of viscous heating.

Heat conduction. Fourier's law gives $\mathbf{q} = -\kappa \nabla T$. For a calorically perfect gas, $e = c_v T$, therefore

$$\mathbf{q} = -\frac{\kappa}{c_v} \nabla e. \quad (50)$$

With Eq. (42), the correction is

$$\rho^{n+1} \frac{e^{n+1} - e^*}{\Delta t} = \nabla \cdot \left(\frac{\kappa}{c_v} \Big|_{\text{eff}} \nabla e^{n+1} \right), \quad (\rho E)^{n+1} = \rho^{n+1} \left(e^{n+1} + \frac{1}{2} |\mathbf{U}^{n+1}|^2 \right). \quad (51)$$

The coefficient must be κ/c_v . The OpenFOAM thermophysical library defines $\alpha = \kappa/c_p$. If the solver passes κ/c_p to this Laplacian, the conductive heat flux is too small by the factor γ . The effective Prandtl number is then γPr . Check the coefficient in the v2412 source^[unverified].

Equation of state. The thermophysical model updates T and ψ from e^{n+1} . Then $p^{n+1} = \rho^{n+1} \psi^{n+1}$.

A.7 Discrete conservation

Sum Eq. (19) over all cells. Each internal face occurs twice with opposite signs $s_{P,f}$. Only the boundary faces remain.

Mass. $\sum_P V_P \rho_P^{n+1} = \sum_P V_P \rho_P^n - \Delta t \sum_{f \in \partial \Omega} F_f^\rho$.

Momentum. The predictor is in conservation form. In the correction, $\sum_P V_P \rho_P^{n+1} (\mathbf{U}_P^{n+1} - \mathbf{U}_P^*)$ equals Δt times a sum of face fluxes of $\mu_{\text{eff}} \nabla \mathbf{U}$ and $\boldsymbol{\tau}_{\text{MC}}$. The internal faces cancel.

Total energy. The predictor is in conservation form. The kinetic-energy term in e^* and in $(\rho E)^{n+1}$ uses the same velocity $\mathbf{U}^{n+1}$. Therefore

$$(\rho E)^{n+1} - (\rho E)^* = \rho^{n+1} (e^{n+1} - e^*) = \Delta t \nabla \cdot \left(\frac{\kappa}{c_v} \Big|_{\text{eff}} \nabla e^{n+1} \right), \quad (52)$$

which is a sum of face fluxes. The internal faces cancel.

Mass, momentum and total energy are therefore conserved exactly, to the tolerance of the linear solvers. The splitting error is an error in accuracy, not in conservation. The explicit viscous work and the implicit momentum correction do not use the same velocity. This gives an $O(\Delta t)$ error in the distribution of energy between kinetic and internal forms.

The Lax–Wendroff theorem [6] applies to a scheme in conservation form with a consistent flux. The flux of Eq. (25) is consistent: $H(u, u) = f(u)$. If the numerical solution converges under mesh refinement, the limit is a weak solution. A weak solution satisfies the Rankine–Hugoniot conditions, which set the shock speed and the jumps. The theorem does not select the entropy solution. The numerical diffusion ω_f selects it in practice, but no proof exists for the limited scheme on unstructured meshes.

A.8 Time-step restriction

The face coefficients of Eq. (31) are the largest signal fluxes through face f . The code defines^[unverified]

$$\text{amaxSf} = \max(|\alpha \phi_f^+ - \omega_f|, |(1 - \alpha) \phi_f^- + \omega_f|), \quad (53)$$

and

$$\text{Co} = \frac{\Delta t}{2} \max_P \left(\frac{1}{V_P} \sum_{f \in \partial P} \text{amaxSf}_f \right). \quad (54)$$

Check on a uniform 1D mesh with $|\mathbf{S}_f| = A$ and $V_P = A \Delta x$: if the maximum wave speed is a on both faces of the cell, $\sum_f \text{amaxSf}_f / V_P = 2a / \Delta x$. Then $\text{Co} = a \Delta t / \Delta x$, the classical Courant number. The factor $\frac{1}{2}$ compensates for the two faces.

With `adjustTimeStep yes`, the standard OpenFOAM time-step control sets

$$\Delta t^{n+1} = \min \left(\min [\lambda, 1 + 0.1 \lambda, 1.2] \Delta t^n, \text{maxDeltaT} \right), \quad \lambda = \frac{\text{maxCo}}{\text{Co}}. \quad (55)$$

The time step decreases immediately when Co is too large. It increases by 20 % per step at most.

A.9 Algorithm summary

Algorithm 1 gives one time step. Code variable names are in typewriter font.

Algorithm 1 One time step of rhoCentralFoam.

Require: $\rho^n, (\rho\mathbf{U})^n, (\rho E)^n, \mathbf{U}^n, e^n, \psi^n, \Delta t^n$

Face states and coefficients

- 1: Reconstruct $\rho^\pm, (\rho\mathbf{U})^\pm, (1/\psi)^\pm, e^\pm, c^\pm$ with the limiter (§A.4)
- 2: $\mathbf{U}^\pm \leftarrow (\rho\mathbf{U})^\pm / \rho^\pm; \quad p^\pm \leftarrow \rho^\pm (1/\psi)^\pm; \quad \phi^\pm \leftarrow \mathbf{U}^\pm \cdot \mathbf{S}_f$
- 3: Compute $a^\pm, \alpha, \omega$ (Kurganov or Tadmor) (§A.3)
- 4: $\text{aphiv_pos} \leftarrow \alpha\phi^+ - \omega; \quad \text{aphiv_neg} \leftarrow (1 - \alpha)\phi^- + \omega$

Time step

- 5: Compute Co and set Δt^{n+1} (§A.8); $t \leftarrow t + \Delta t$

Inviscid fluxes

- 6: $F^\rho \leftarrow \text{aphiv_pos} \rho^+ + \text{aphiv_neg} \rho^-$
- 7: $\mathbf{F}^{\rho\mathbf{U}} \leftarrow \text{aphiv_pos} (\rho\mathbf{U})^+ + \text{aphiv_neg} (\rho\mathbf{U})^- + [\alpha p^+ + (1 - \alpha)p^-] \mathbf{S}_f$
- 8: $F^{\rho E} \leftarrow \text{aphiv_pos} (\rho H)^+ + \text{aphiv_neg} (\rho H)^- + \omega (p^+ - p^-)$

Mass

- 9: $\rho^{n+1} \leftarrow \rho^n - \Delta t V^{-1} \sum_f s F^\rho$

Momentum

- 10: $(\rho\mathbf{U})^* \leftarrow (\rho\mathbf{U})^n - \Delta t V^{-1} \sum_f s \mathbf{F}^{\rho\mathbf{U}}; \quad \mathbf{U}^* \leftarrow (\rho\mathbf{U})^* / \rho^{n+1}$
- 11: **if** viscous **then**
- 12: $\boldsymbol{\tau}_{\text{MC}} \leftarrow \mu_{\text{eff}} \text{dev}_2[(\nabla\mathbf{U})^T]$
- 13: Solve $\rho^{n+1}(\mathbf{U}^{n+1} - \mathbf{U}^*) / \Delta t - \nabla \cdot (\mu_{\text{eff}} \nabla \mathbf{U}^{n+1}) = \nabla \cdot \boldsymbol{\tau}_{\text{MC}}$
- 14: $(\rho\mathbf{U})^{n+1} \leftarrow \rho^{n+1} \mathbf{U}^{n+1}$
- 15: **end if**

Energy

- 16: $(\rho E)^* \leftarrow (\rho E)^n - \Delta t V^{-1} \sum_f s [F^{\rho E} - \text{sigmaDotU}]$
- 17: $e^* \leftarrow (\rho E)^* / \rho^{n+1} - \frac{1}{2} |\mathbf{U}^{n+1}|^2; \quad \text{update thermophysical model}$
- 18: **if** viscous **then**
- 19: Solve $\rho^{n+1}(e^{n+1} - e^*) / \Delta t - \nabla \cdot (\alpha_{\text{eff}} \nabla e^{n+1}) = 0; \quad \text{update thermophysical model}$
- 20: $(\rho E)^{n+1} \leftarrow \rho^{n+1} (e^{n+1} + \frac{1}{2} |\mathbf{U}^{n+1}|^2)$
- 21: **end if**

Closure

- 22: $p^{n+1} \leftarrow \rho^{n+1} / \psi^{n+1}; \quad \text{correct boundary values; correct turbulence model}$
-

References

- [1] C. J. Greenshields, H. G. Weller, L. Gasparini, J. M. Reese. Implementation of semi-discrete, non-staggered central schemes in a colocated, polyhedral, finite volume framework, for high-speed viscous flows. *Int. J. Numer. Methods Fluids* 63(1):1–21, 2010. doi:10.1002/fld.2069.
- [2] A. Kurganov, E. Tadmor. New high-resolution central schemes for nonlinear conservation laws and convection–diffusion equations. *J. Comput. Phys.* 160(1):241–282, 2000. doi:10.1006/jcph.2000.6459.
- [3] A. Kurganov, S. Noelle, G. Petrova. Semidiscrete central-upwind schemes for hyperbolic conservation laws and Hamilton–Jacobi equations. *SIAM J. Sci. Comput.* 23(3):707–740, 2001. doi:10.1137/S1064827500373413.
- [4] P. K. Sweby. High resolution schemes using flux limiters for hyperbolic conservation laws. *SIAM J. Numer. Anal.* 21(5):995–1011, 1984. doi:10.1137/0721062.
- [5] A. Harten, P. D. Lax, B. van Leer. On upstream differencing and Godunov-type schemes for hyperbolic conservation laws. *SIAM Rev.* 25(1):35–61, 1983. doi:10.1137/1025002.
- [6] P. D. Lax, B. Wendroff. Systems of conservation laws. *Commun. Pure Appl. Math.* 13(2):217–237, 1960. doi:10.1002/cpa.3160130205.